WORKED SOLUTION

STEP I 1990, Question 9

Pure • Coordinate geometry

STEP I 1990, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q9

The key idea

The curve $y = 1/x$ bends upwards (it is convex), so every tangent lies below it. The two trapezia under the tangents therefore fit inside the region under the curve between $x = 1$ and $x = b$, and that region has area $\ln b$. Comparing the two areas gives the lower bound for the logarithm. A rectangle that sits above the curve gives the upper bound in the same way.

Solution

The point C

For $y = 1/x$ we have $\frac{\mathrm{d}y}{\mathrm{d}x} = -1/x^2$. So the tangent at $(t, 1/t)$ is

$$y - \frac{1}{t} = -\frac{1}{t^2}(x - t), \qquad \text{that is} \qquad y = \frac{2}{t} - \frac{x}{t^2}.$$

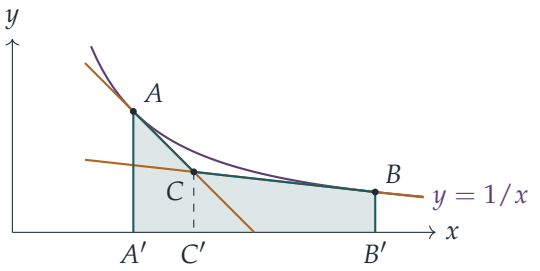
With $t = 1$ the tangent at A is $y = 2 - x$. With $t = b$ the tangent at B is $y = \frac{2}{b} - \frac{x}{b^2}$. Where they meet,

$$2 - x = \frac{2}{b} - \frac{x}{b^2} \quad \Longrightarrow \quad x\left(1 - \frac{1}{b^2}\right) = 2\left(1 - \frac{1}{b}\right).$$

Since $b > 1$ we may divide by $1 - \frac{1}{b} \neq 0$, which leaves $x\left(1 + \frac{1}{b}\right) = 2$. Hence

$$C = \left(\frac{2b}{b+1}, \frac{2}{b+1}\right),$$

where the y-coordinate comes from $y = 2 - x$. Note that $1 < \frac{2b}{b+1} < b$: the left inequality is $b + 1 < 2b$, and the right one is $2 < b + 1$. Both hold because $b > 1$, so C lies between A and B horizontally.



The figure is drawn for $b = 3$. The shaded region is made up of the two quadrilaterals.

The sum of the areas

Each quadrilateral has two vertical sides, so it is a trapezium, and its area is the width times the mean of the two vertical sides.

For $ACC'A'$ the width is $\frac{2b}{b+1} - 1 = \frac{b-1}{b+1}$, and the vertical sides are 1 and $\frac{2}{b+1}$:

$$\text{area}(ACC'A') = \frac{1}{2} \cdot \frac{b-1}{b+1}\left(1 + \frac{2}{b+1}\right) = \frac{(b-1)(b+3)}{2(b+1)^2}.$$

For $CBB'C'$ the width is $b - \frac{2b}{b+1} = \frac{b(b-1)}{b+1}$, and the vertical sides are $\frac{2}{b+1}$ and $\frac{1}{b}$:

$$\text{area}(CBB'C') = \frac{1}{2} \cdot \frac{b(b-1)}{b+1}\left(\frac{2}{b+1} + \frac{1}{b}\right) = \frac{(b-1)(3b+1)}{2(b+1)^2}.$$

Adding, the numerators combine to $(b-1)(4b+4) = 4(b-1)(b+1)$, so

$$\text{area}(ACC'A') + \text{area}(CBB'C') = \frac{2(b-1)}{b+1}.$$

The inequalities

? **Ask yourself** Why must the two trapezia fit under the curve?

For $x > 0$, the gap between the curve and the tangent at $(t, 1/t)$ is

$$\frac{1}{x} - \left(\frac{2}{t} - \frac{x}{t^2}\right) = \frac{t^2 - 2tx + x^2}{t^2 x} = \frac{(x-t)^2}{t^2 x} \geq 0.$$

So each tangent lies on or below the curve for all $x > 0$. The top edge of the shaded region is the tangent at A on $[1, \frac{2b}{b+1}]$ and the tangent at B on $[\frac{2b}{b+1}, b]$. Both pieces lie below $y = 1/x$, and the region sits above the x-axis. So the shaded region is contained in the region under the curve from $x = 1$ to $x = b$, and

$$\frac{2(b-1)}{b+1} \leq \int_1^b \frac{\mathrm{d}x}{x} = \ln b.$$

For the upper bound, $\frac{1}{x} \le 1$ for $x \ge 1$, so the region under the curve lies inside the rectangle with base $A'B'$ and height 1. Hence

$$\ln b = \int_1^b \frac{\mathrm{d}x}{x} \le \int_1^b 1\,\mathrm{d}x = b - 1.$$

Now let $z > 0$ and put $b = 1 + z$, so that $b > 1$ as required. Then $b - 1 = z$ and $b + 1 = 2 + z$, and the two inequalities together read

$$\frac{2z}{2+z} \le \ln(1+z) \le z,$$

which is what we had to prove. (Both inequalities are in fact strict for $z > 0$, because the curve and the bounding lines meet only at isolated points.)

Checks

- With $b = 1$, C would be $(1, 1) = A$ and the area sum would be 0, as it should.
- With $b = 2$: $C = (\frac{4}{3}, \frac{2}{3})$, and the area sum is $\frac{2}{3} \approx 0.667$, just below $\ln 2 \approx 0.693$, which is in turn below 1.
- For small z, $\frac{2z}{2+z} = z - \frac{z^2}{2} + \cdots$ agrees with $\ln(1+z) = z - \frac{z^2}{2} + \cdots$ up to the z^2 term, so the lower bound is very sharp near 0.

We checked the coordinates and both trapezium areas in exact arithmetic for several values of b, and the inequality for 10^5 random z between 10^{-6} and 10^6.

Common slips

- Forgetting to check that C lies between $x = 1$ and $x = b$. Without this, the "quadrilaterals" need not be the shapes you think.
- Saying that the tangents are below the curve "from the picture". The one-line calculation with $(x - t)^2$ is what makes it a proof.

Beyond STEP

Comparing a function with simple shapes above and below its graph is the idea behind Riemann sums, and it is how integrals are rigorously defined in a first analysis course. The fact that tangents of a convex function lie below its graph is the start of the theory of convex functions. It leads to Jensen's inequality and to many classical inequalities, including AM–GM. Abbott's *Understanding Analysis* develops the integral in exactly this spirit.

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